Practical 9:

Aim: system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

```
GNU nano 8.7                                    optimal.c
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frame[3], n=12, frames=3;
    int i,j,k,hit=0,fault=0,pos,far;

    for(i=0;i<frames;i++)
        frame[i] = -1;

    for(i=0;i<n;i++) {
        int found = 0;

        for(j=0;j<frames;j++) {
            if(frame[j] == pages[i]) {
                hit++;
                found = 1;
                break;
            }
        }

        if(!found) {
            fault++;

            for(j=0;j<frames;j++) {
                if(frame[j] == -1) {
                    frame[j] = pages[i];
                    found = 1;
                    break;
                }
            }

            if(!found) {
                far = -1;
                pos = -1;

                for(j=0;j<frames;j++) {
                    int next = -1;

                    for(k=i+1;k<n;k++) {
                        if(frame[j] == pages[k]) {
                            next = k;
                            break;
                        }
                    }
^G Help      ^O Write Out   ^F Where Is    ^K Cut       ^T Execute   ^C Location   M-U Undo     M-A Set Mark   M-] To Bracket
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo     M-6 Copy       ^B Where Was
```

```
GNU nano 8.7                                        optimal.c

            for(j=0;j<frames;j++) {
                if(frame[j] == -1) {
                    frame[j] = pages[i];
                    found = 1;
                    break;
                }
            }

            if(!found) {
                far = -1;
                pos = -1;

                for(j=0;j<frames;j++) {
                    int next = -1;

                    for(k=i+1;k<n;k++) {
                        if(frame[j] == pages[k]) {
                            next = k;
                            break;
                        }
                    }

                    if(next == -1) {
                        pos = j;
                        break;
                    }

                    if(next > far) {
                        far = next;
                        pos = j;
                    }
                }

                frame[pos] = pages[i];
            }
        }
    }

    printf("Hits = %d\nFaults = %d\n", hit, fault);
    printf("Hit %% = %.2f\nFault %% = %.2f\n",(float)hit/n*100,(float)fault/n*100);

    return 0;
}

^G Help      ^O Write Out   ^F Where Is    ^K Cut       ^T Execute   ^C Location   M-U Undo     M-A Set Mark   M-] To Bracket
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo     M-6 Copy       ^B Where Was
```

```
FOIN@LAPTOP-MC73RDEH MSYS ~
$ nano optimal.c

FOIN@LAPTOP-MC73RDEH MSYS ~
$ gcc optimal.c -o optimal

FOIN@LAPTOP-MC73RDEH MSYS ~
$ ./optimal
Hits = 5
Faults = 7
Hit % = 41.67
Fault % = 58.33

FOIN@LAPTOP-MC73RDEH MSYS ~
$
```